

Exercise 5 : NULL Functions

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① Employees

SELECT employee_id,
name,
IFNULL(salary, 0) AS salary_with_default
FROM employees;

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

② SELECT employee_id, name,

IFNULL(department, 'Not assigned') AS department name

FROM employees;

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

③ Orders

```
SELECT order_id,  
       customer_id  
FROM   orders  
WHERE  ISNULL(customer_id);
```

order_id	customer_id
103	NULL

④ SELECT order_id,
 customer_id,
 IFNULL(delivery_date, 'Pending') AS delivery_status
FROM orders;

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

⑤ STUDENTS

```
SELECT student_id,  
       name,  
       IFNULL(grade, 0) AS final_grade  
FROM   students;
```

student_id	name	final_grade
1	Ethan	85
2	Mayer	0
3	Olivia	90

⑥ SELECT COUNT(*) AS not_graded_count
 FROM students
 WHERE ISNULL(grade);

not_graded_count
1

⑦ SELECT product_id,
 name,
 price - IFNULL(discount, 0) AS final_price
 FROM products;

product_id	name	Final_price
501	Keyboard	25
502	mouse	10
503	Monitor	100

⑧ Customers

SELECT COUNT(*) AS missing_email_count
 FROM customers
 WHERE ISNULL(email);

missing_email_count
2

⑨ SELECT customer_id,
 name,
 IFNULL(email, No Email) AS email
 FROM customers;

customer_id	name	email
1	Liada	No email
2	Joseph	joseph@mail.com
3	Nia	No email

(10) Payments

```
SELECT payment_id  
       IFNULL(method, 'Unknown') AS payment_method,  
       status  
FROM payments;
```

Payment_id	Payment_method	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

(11) Inventory

```
SELECT item_id,  
       item_name,  
       IFNULL(quantity, 0) AS quantity_check  
FROM inventory;
```

item_id	item_name	quantity_check
1	Pen	0
2	Notebook	150
3	Eraser	0

(12) Employees Extra

```
SELECT emp_id,  
       COALESCE(bonus, commission) AS first_available_reward  
FROM employees_extra;
```

emp_id	first_available_reward
1	300
2	100
3	NULL

13) Classes

SELECT COUNT (*) AS no_room_count
FROM classes
WHERE ISNULL (room);

no_room_count
2

14) Attendance

SELECT student_id,
date,
IFNULL (status, Not Marked) AS attendance_status
FROM attendance;

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

15) Bank Accounts

SELECT account_id,
IFNULL (account_type, 'Unknown') AS account_type,
IFNULL (balance, 0) AS balance
FROM bank_accounts;

account_id	account_type	balance
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

16) Projects

```
SELECT project_id,  
       title,  
       IFNULL(start_date, 'TBD') AS start_display  
FROM projects;
```

Project_id	title	start_display
1	Website revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

18) Suppliers

```
SELECT supplier_id,  
       name,  
       COALESCE(phone, alt_phone, 'No Contact') AS contact_number  
FROM suppliers;
```

Supplier-id	name	contact_number
1	Global Grocks	123456789
2	Best Supplies	987654321
3	ValueSource	No Contact.

17) Reviews

```
SELECT review_id,  
       product_id,  
       IFNULL(comment, 'No Comment'),  
       IFNULL(rating, 0)  
FROM reviews;
```

review	product_id	Comment	rating
1	501	Great product	4
2	502	No Comment	0
3	503	Works fine	3

19) User Settings

```
SELECT user_id,  
       IFNULL(theme, 'light') AS theme_set,  
       IFNULL(language, 'English') AS language_set,  
       IFNULL(timezone, 'UCT') AS timezone_set.
```

FROM user_settings;

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC
3	Light	English	UTC

20) Maintenance

```
SELECT record_id,  
       machine_id,  
       IFNULL(issue, 'Unknown Issue') AS issue_log,  
       IFNULL(technician, 'Not Assigned') AS tech_name
```

FROM maintenance;

record_id	machine_id	issue_log	tech_name
1	m101	Overheating	Not Assigned
2	m102	Unknown_issue	Not Assigned
3	m103	Jammed	Alex